Activity 4.2 – Pipeline Flush

1.Design Approach:

Designed a pipeline flush module using Verilog HDL to clear unwanted instructions during reset or branch-related flush conditions. Implemented logic to insert a NOP instruction into the pipeline.

2.Verification Steps:

- Created a separate testbench.
- Applied reset.
- Provided sample instruction inputs.
- Activated the flush signal.
- Checked NOP instruction generation.
- Verified normal instruction transfer.
- Simulated using ModelSim.

3.Results:

The testbench checks instruction transfer and NOP insertion during flush conditions. PASS/FAIL messages are displayed.

4.Issues Faced:

Flush signals must be activated at the correct time. Reset and flush priority must be defined properly. NOP instruction encoding must be accurate.

5.Conclusion:

Successfully designed and verified a pipeline flush module using Verilog HDL.